

Mohammad Hossein Namdar

Department of Physics, Shahid Beheshti University, Tehran, Iran

EDUCATION

Shahid Beheshti University, Tehran, Iran

Sep 2021 – Currently

- M.Sc. Major in Physics - Gravity & Cosmology
- Supervisor: Dr. Nima Khosravi
- Advisor: Dr. Farnik Nikakhtar
- GPA: 18.19/20.00
- Ranked 1st out of 14 people

Isfahan University of Technology, Isfahan, Iran

Sep 2018 – Sep 2021

- B.Sc. Major in Physics
- GPA: 18.20/20.00 (Special Courses GPA: 19.07/20.00)
- Ranked 1st out of 53 people

RESEARCH INTERESTS

Cosmology, Astrophysics, Large Scale Structure Formation, Modified Gravity Theories (Dark Energy & Dark Matter), Optimal Transport Theory & Application to Physics such as Modify Gravity, Computational Physics, Simulation, Data Analysis

RESEARCH PROJECTS

Monge-Ampère Gravity at Cosmological Scales

- **Supervisors:** Dr. Nima Khosravi, Dr. Farnik Nikakhtar and Prof. Roya Mohayaee
- **Summary:** This is my master's thesis in which we investigated the merger history of dark matter halos within the Excursion Set Theory framework, utilizing the Monge-Ampère Gravity (MAG) instead of Poisson Gravity (PG). MAG is a non-linear modification of PG that is predicted by statistical physics and capturing Anisotropic Collapse for cosmic structure. We determined the redshifts at which the trajectories passed through the barrier of critical density that was determined in the Gravitational Collapse after which they collapsed to form virialized dark matter halos. Consequently, we evaluated the statistical properties of dark matter halos, including Growth Rate, Mass Function, and Merger History, within this MAG and compared them with the results obtained from the PG. Our research covers multiple topics in the fields of Physics, Mathematics, and Computer Science apart from Cosmology, such as Stochastic Methods, Numerical methods, Nonlinear Dynamics and Programming with Python and Mathematica.

COURSE PROJECTS

Quantum Cosmology and the Calculation of the Wheeler–DeWitt equation

- **Course:** Quantum Mechanics II
- **Supervisor:** Dr. Mehdi Abdi

Cosmic Ray Experiments for Neutrino Mass Studies

- **Course:** Introduction to Elementary Particles
- **Supervisor:** Dr. Soroush Shakeri

Ising Model Enhanced with Monte Carlo and Genetic Algorithms

- **Course:** Foundation of Numerical Simulations
- **Supervisor:** Dr. Ali Hosseiny

Simulation of Molecular Dynamics

- **Course:** Biophysics
- **Supervisor:** Dr. Mehdi Alaei

TECHNICAL SKILLS

Programming Languages :

Advanced: C/C++, Python

Intermediate: VisualBasic 6.0, QBasic, HTML

Mathematical Programs :

Advanced: Maple

Intermediate: Mathematica, MATLAB

Simulation Tools :

Advanced: Proteus, CodeVisionAVR

Basic Familiarity: xAct, CAMB, CosmoMC, GADGET

Documentation & Software :

Advanced: L^AT_EX, Microsoft Office, Ubuntu (Linux), Origin

SELECTED COURSEWORK & GRADES	Graduate Courses	
	Gravitation II (19.7/20), Cosmology II (17.5/20), Cosmology I (18/20), Astrophysics (Audit), Advance Statistical Mechanics I (19.7/20), Advance Quantum Mechanics (17.1/20), Foundation of Numerical Simulations (20/20), Computational Physics [with <u>Machine Learning</u>] (19.5/20), Special Topics in Cosmology (Audit)	
	Undergraduate Courses	
	Introduction to Elementary Particles (20/20), Quantum Mechanics II (19.01/20), Electromagnetics II (17.5/20), Analytical Mechanics I (20/20), Statistical Mechanics (19.8/20), Mathematical Physics I,II (19/20), Differential Equations (20/20), Thermodynamics and Statistical Physics (17.75/20), Biophysics (19.8/20)	
TEACHING EXPERIENCE	Teacher Assistant at Shahid Beheshti University	
	▪ Gravitation I (GR, BHs, GW, ...)	Jan 2023 – Jul 2023
	<i>Dr. Nima Khosravi</i>	
	Teacher Assistant at Isfahan University of Technology	
	▪ Analytical Mechanics II	Sep 2021 – Dec 2021
	<i>Dr. Soroush Shakeri</i>	
	▪ Physics of Agriculture and Natural Resources	Jan 2021 – Jul 2021
	<i>Dr. Soroush Shakeri</i>	
	▪ Electronics Lab	Dec 2019 – Feb 2020
	<i>Dr. Mohammad Ahangarian</i>	
	Teacher at Allameh Helli 1 Highschool	
	Olympiad question design, Spherical Astronomy and Sky Observations	Jun 2016 – Jun 2017
	Teacher at Rahyar Highschool	
	Cosmology, Dynamics of Galaxies, Astronomy and Sky Observations	Jun 2023 – Aug 2023
TALKS & POSTERS	Monge-Ampère Gravity at Cosmological Scales	
	– Held at ICTP, Trieste, Italy	Jun 2024
	Astronomy and Cosmology: From Theory to Observation	
	– Held at Isfahan University of Technology, Physic Department Students Scientific Association, Isfahan, Iran (IN-PERSON)	Nov 2018
	– summary: <i>In this talk, I discussed the historical development of observational astronomy, scientists' efforts to understand the universe's structure and galaxies throughout history, and briefly touched on topics like black holes and proposed geometric theories.</i>	
SCHOOLS & WORKSHOPS	Attended:	
	▪ ICTP's Summer School on Cosmology	Jun 2024
	– Held at ICTP, Trieste, Italy (IN-PERSON)	
	↔ <i>Accepted as Full-Funded participant among candidates.</i>	
	▪ Cosmology: From Theory to Observation	Jul 2023
	– Held at IPM Institute for Research in Fundamental Sciences, Tehran, Iran (IN-PERSON)	
	▪ 25th Meeting on Research in Astronomy at IASBS	Jun 2023
	– Held at Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan, Iran (IN-PERSON)	
	▪ Formation, Evolution and Structure of Galaxies	Oct 2021
	– Held at Iranian National Observatory, Tehran, Iran (VIRTUALLY)	
	▪ National School on Statistical Analysis of Cosmological Random Field	Sep 2021
	– Held at Shahid Beheshti University, Tehran, Iran (VIRTUALLY)	
	▪ NBIA Summer School on Gravitational Wave Astrophysics	Aug 2021
	– Held at Niels Bohr Institute, Denmark (VIRTUALLY)	
	Organized:	
	▪ STAR WORKSHOP	Aug 2022 - Feb 2023
	– Held at Shahid Beheshti University, Physic Department Students Scientific Association, Tehran, Iran (VIRTUALLY)	
	– summary: <i>I led a workshop encompassing theoretical and software components, where I taught advanced astronomy topics and performed data analysis using Starry Night software.</i>	
	▪ Workshop for Teaching and Solving Physics Problems for the Exam	Feb 2023
	– Held at Shahid Beheshti University & Amirkabir University of Technology, Physic Department Students' Scientific Association, Tehran, Iran (VIRTUALLY)	

HONORS & AWARDS

Ranked Top :

- Ranked 16th in National Physics Olympiad among Iranian Undergraduate Students, Iran, 2021
 ↪ *Ranked 1 among 9 regions out of 10 regions.*
- Ranked 18th in Nationwide Graduate School Entrance Exam for M.Sc. in physics amongst more than 30600 participants, Iran, 2021
- Ranked 4th in the Messier marathon, Iran, 2016
 ↪ *A Sky Observation tournament held by Nojum magazine.*
- Semifinalist in National Astronomy and Astrophysics Olympiad, Young Scholars Club, Iran, 2017
- Ranked 7th among 150 teams in "HelliNet" programming competitions under 18 years old (with C++ language), Iran, 2014

Full Fund Scholarships & Memberships:

- Graduated B.Sc. with **honor** -Achieved to enroll in the M.Sc. level without any entrance examination.
- Member of the National Elite Foundation (since 2018)
 ↪ *Iran's National Elites Foundation (INEF) is a statewide organization, composed of members with significant scientific and executive background.*
- Member of the National Organization for Development of Exceptional Talent (NODET)

Others :

- 4 times Winner of the best and most creative physics problem solver of Isfahan House of Mathematics among physics students in the country, Isfahan, Iran, 2018-2020
- I was selected for the university math team for the International Mathematics Competition for University (IMC)
 ↪ *Unfortunately, I was not able to attend this program due to the Covid-19 pandemic.*

PROFESSIONAL MEMBERSHIPS

President of Astronomy group

Sep 2022 – Sep 2023

- Shahid Beheshti University, Tehran, Iran
 I established and led the astronomy group at Shahid Beheshti University, Organizing stargazing events and Observation trips, and managing teams for exhibitions and workshops in the physics department for one year.

Member of the Physics Society of Iran

May 2021 – Currently

I work within the Iranian Physics Association, focusing on Cosmology, Astronomy, Particle Physics and Complex Systems. This association, overseen by physics professors, offers conferences and internships for its members.

Member of Scientific Society of Astronomy

Sep 2018 – Nov 2018

- Isfahan University of Technology, Isfahan, Iran
 I briefly served in the Faculty Astronomy Association, conducting Conferences, Teaching Courses, and Participating in Observation programs like "Explaining the Sky in Planetarium." I also served as the editor-in-chief of Einsteinium magazine.

OTHER ACADEMIC ACTIVITIES

Magazine

- The Author of the Cosmology section of the 32nd edition of "**Tekaneh**", Sharif University of Technology, Mar 2023
- I led the Astronomy and Cosmology section at "**Quark**" Journal of Shahid Beheshti University, overseeing content revision for four individuals, and suggesting topics for their articles, Winter 2023

Open Day and Harekat Festival (Performing physics experiments for students and the public)

- Shahid Beheshti University, Tehran, Iran, 2023
- Isfahan University of Technology, Isfahan, Iran, 2018

LANGUAGES

Persian: Native Speaker

English: Excellent